LAB 3 Individual Deliverable 2

Modelling and Control of UAVs

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Scope. This report answers the two individual modelling and control questions in MIT VNAV Lab 3. It analyzes the rank of the quadrotor control-allocation matrix after reversing the rotation directions of propellers 3 and 4, and then explains model-based compensation of body-frame quadratic drag in a geometric controller.

Important convention. The signs below follow the simulator convention shown in the Lab 3 materials. If the course lecture notes use a different row or axis ordering, the same rank argument remains valid after consistently reordering or changing signs.

1. Problem statement and assumptions

Let the body-frame positions of the four rotors be arranged in an X configuration:

```
rho_1 = ( l,  l, 0)       rho_2 = (-l,  l, 0)
rho_3 = (-l, -l, 0)       rho_4 = ( l, -l, 0)
where l = d / sqrt(2), and d is the centre-to-rotor distance.
```

For rotor i, define u_i = omega_i |omega_i|. For positive commanded speeds, u_i = omega_i^2. The lift is f_i = c_f u_i, and c_d is the magnitude of the aerodynamic drag-torque coefficient. The body z-axis is taken upward, consistent with the ROS/simulator convention.

2. Question 1 - Quadrotor structure and controllability

2.1 Normal quadrotor allocation matrix

The force from rotor i is f_i e_3, with e_3 = (0, 0, 1)^T. Therefore, the moment produced by the thrust is:

```
rho_i x (f_i e_3) = [ y_i f_i,  -x_i f_i,  0 ]^T.
```

Using W = [F_z, tau_x, tau_y, tau_z]^T and u = [u_1,u_2,u_3,u_4]^T, the normal vehicle has:

```
W = F_normal u

F_normal =
[  c_f       c_f        c_f        c_f      ]
[  l c_f    l c_f     -l c_f     -l c_f    ]
[ -l c_f    l c_f      l c_f     -l c_f    ]
[  c_d      -c_d       c_d       -c_d     ].
```

The rows represent total upward force, roll torque, pitch torque, and yaw drag torque. The yaw signs are (+,-,+,-) for the normal rotor set.

```
After removing nonzero row scale factors,

F_tilde_normal =
[  1   1   1   1 ]
[  1   1  -1  -1 ]
[ -1   1   1  -1 ]
[  1  -1   1  -1 ].

det(F_tilde_normal) = -16, so rank(F_normal) = 4.
```

The nonzero determinant proves that the four wrench components are locally independent control directions for the normal quadrotor, assuming c_f, c_d, and l are nonzero.

2.2 Modified quadrotor matrix

When the rotation directions of propellers 3 and 4 are reversed, their thrust positions and lift contributions remain unchanged under the simulator convention, while their yaw drag signs reverse. The yaw row changes from (+,-,+,-) to (+,-,-,+):

```
F_modified =
[  c_f       c_f        c_f        c_f      ]
[  l c_f    l c_f     -l c_f     -l c_f    ]
[ -l c_f    l c_f      l c_f     -l c_f    ]
[  c_d      -c_d      -c_d        c_d     ].

F_tilde_modified =
[  1   1   1   1 ]
[  1   1  -1  -1 ]
[ -1   1   1  -1 ]
[  1  -1  -1   1 ].

The fourth row equals the negative of the third row.
Therefore det(F_tilde_modified) = 0 and rank(F_modified) = 3.
```

2.3 Why [x,y,z,Psi] tracking is impossible in general

A quadrotor tracks three-dimensional position by producing a desired total force and orienting the body so that its thrust vector has the required horizontal components. At the same time, the yaw angle Psi requires an independent yaw torque. The required independent wrench directions are therefore F_z, tau_x, tau_y, and tau_z.

For the normal vehicle, rank(F_normal)=4, so the allocation map can independently assign the four wrench components, within actuator limits. For the modified vehicle, rank(F_modified)=3. Its achievable wrench vectors lie in only a three-dimensional subspace. In particular, the yaw torque is linearly coupled to the pitch-related row, so a rotor-speed change intended to create yaw also changes another required wrench component.

Therefore the modified vehicle is underactuated for the general task of independently tracking [x,y,z,Psi]. It may track restricted motions or maintain some states, but it cannot guarantee simultaneous arbitrary three-dimensional position and yaw tracking.

3. Question 2 - Drag-aware geometric control

3.1 Given body-frame drag model

The given empirical model is:

```
F^b = D (v^b)^2,

D = [ 0.1   0     0   ]
    [ 0     0.1   0   ]
    [ 0     0     0.2 ],

(v^b)^2 = [ -v_x^b |v_x^b|,
             -v_y^b |v_y^b|,
             -v_z^b |v_z^b| ]^T.
```

The signed square makes the force oppose motion. For example, positive v_x^b gives a negative x drag force, while negative v_x^b gives a positive x drag force. The z coefficient is larger than the x and y coefficients, so the same speed magnitude produces stronger modelled drag along body z.

3.2 Transforming drag into the world frame

The drag model is expressed in the body frame, whereas the translational controller normally uses world-frame position, velocity, acceleration, and force. Let R_wb map a body-frame vector into the world frame. Then:

```
v^b = (R_wb)^T v^w

Fhat^b = D [ -v_x^b |v_x^b|,
             -v_y^b |v_y^b|,
             -v_z^b |v_z^b| ]^T

Fhat^w = R_wb Fhat^b.
```

The rotation is necessary because the body axes rotate with the UAV. Adding Fhat^b directly to a world-frame force equation would mix coordinates and produce an incorrect compensation direction.

3.3 Feed-forward compensation

Let F_base^w be the nominal force from the geometric controller. It includes position error, velocity error, desired acceleration, and gravity compensation. With drag included, the translational dynamics are:

```
m x_ddot = F_rotor^w + F_drag^w + m g^w.
```

The rotor force should cancel the estimated disturbance while still producing the nominal desired acceleration. Therefore the compensated desired force is:

```
F_des^w = F_base^w - Fhat^w.
```

The minus sign is essential: Fhat^w is the force exerted by the air on the vehicle, so the rotor-generated force must oppose it. The corrected force is then used to define the desired body z-axis:

```
b_3d = F_des^w / ||F_des^w||.
```

Combining b_3d with the desired yaw direction defines the desired attitude R_d. The controller then computes the desired thrust and attitude moment, and finally converts the desired wrench to rotor speeds through the allocation matrix.

3.4 Implementation sequence

Step	Operation
1	Read the current world-frame velocity and attitude.
2	Transform velocity into the body frame using (R_wb)^T.

Step	Operation
3	Evaluate the signed quadratic drag model to obtain Fhat^b.
4	Transform Fhat^b back to the world frame using R_wb.
5	Subtract the estimated world-frame drag from the nominal desired force.
6	Recompute desired attitude, thrust, and desired moment.
7	Apply wrench-to-rotor allocation and send rotor angular velocities.

3.5 Expected improvements and limitations

- The compensation should reduce position lag and radius error during fast circular motion.
- It should improve velocity tracking during acceleration and deceleration.
- It should reduce performance differences between directions with different drag coefficients.
- The larger z coefficient means vertical motion may require a larger correction for the same speed magnitude.
- Velocity noise is amplified by the signed-square model; filtering or a small deadband may be needed near zero velocity.
- Incorrect coefficients can cause under-compensation or over-compensation, so compensation should be bounded by safe actuator limits.

4. Conclusion

The normal quadrotor allocation matrix has rank four. Reversing the rotation directions of propellers 3 and 4 changes the yaw row so that it becomes linearly dependent on the pitch row, reducing the rank to three. The modified quadrotor therefore cannot independently command all four wrench components required for general [x,y,z,Psi] tracking.

For the drag model, velocity is first expressed in the body frame, the signed quadratic drag is evaluated there, and the resulting force is rotated into the world frame. Subtracting the estimated world-frame drag from the nominal desired force provides a model-based feed-forward compensation that can improve high-speed tracking when the model and velocity estimate are reliable.

References

[1] MIT VNAV. Lab 3 Exercises. https://vnav.mit.edu/labs_2023/lab3/exercises.html

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[3] ROS geometry2, tf2_geometry_msgs, noetic-devel. https://github.com/ros/geometry2/tree/noetic-devel/tf2_geometry_msgs